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Capstone Assignment [COA]

CASE STUDY: Cache Design for a online Examination System.

* Introduction:

An online examination system is accessed by thousands of students simultaneously, especially during peak exam time.

Problem addressing:

- Slow response time
- Delay in loading question
- High server load.

Design an efficient memory hierarchy strategy to improve system performance using LOA:

⇒ Concepts:

- Locality of Preference:
  Programs tend to access the same data repeatedly

→ Types:

1. Temporal Locality: Recently accessed data is reused.
   Eg → Same exam question accessed multiple times.
2. Spatial Locality: Nearly data is accessed
   Eg → Loading next questions.

CPU → Cache → Main memory → Disk

- Cache Memory :
  Cache is a small, high-speed memory placed b/w CPU and RAM.

Cache Mapping Technique:

- Direct Mapping :
  Each memory block → only one cache location

  Memory Block → Fixed cache line

- Fully Associate Mapping :

  Any block → any cache location.

  Memory Block → Any Cache Line

- Mapping :

  Cache divided into sets

  Block map to a set, not fixed line

  Memory → Set → Any Line in set

★ Trade off in Cache Design

Cache design is an online examination system involves carefull balance ~~b/w~~ between performance, cost and

and complexity. Increasing the cache size improves the hit rate and reduces memory access time, but it also increases hardware cost and may slightly increase access latency. Similarly a larger block size helps in exploiting spatial locality, allowing more nearby data to be fetched at once. However, it increases the miss penalty and may waste bandwidth if unnecessary data is loaded. Increasing associativity reduces conflict misses and improves cache efficiency, but it makes cache lookup slower and the design more complex.

⇒ Performance Issues:

1) Cache Misses → when required data is not found in cache, the system must access main memory causing delay,
Eg → First time loading a question → miss → Slower response.

2) High miss penalty: when a cache miss occurs, fetching data from RAM takes much more time.
Eg → If exam questions are not cache, each request goes to main memory → delay increases.

3) Memory Stall: CPU has to wait when data is not available immediately.

4) Poor Locality of Reference: If the program does not reuse data efficiently. Cache hit rate decrease and Performance drops.

5) Conflict Miss : In direct mapping, multiple memory blocks map to the same cache line.

★ Recommendations :

To achieve high performance in an online examination system with heavy concurrent users, an optimized cache design must be implemented using COA principles. The system should adopt a multi-level cache hierarchy [L1, L2, L3] to minize memory access time and reduce CPU ~~ts~~ waiting. Amount mapping techniques, set associative mapping is recommended as it provides a balanced trade-off b/w speed, cost and miss rate.

Further, a write back cache policy with write-allocate should be used to improve performance by reducing unnecessary memory writes. Also, maintaining a high cache hit rate is essential to ensure low Average Memory Access time. Overall combining efficient cache hierarchy, optimized mapping and proper write policy will significantly improve system responsiveness, reduce latency and ensure smooth execution of the online examination system under heavy load.

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CASE STUDY 2: Storage Planning for a Video Surveillance System

1) Role of Secondary Storage
Secondary storage is used to store large volumes of video data continuously. It provide high capacity, long-term storage, and ensures recorded footage can be retrieved later for security analysis.

2) Magnetic Disks vs Flash Memory
- Magnetic Disk (HDD):
  → High storage capacity
  → low cost per kB
  → Slower access speed

- Flash Memory (SSD):
  → Faster data access
  → More reliable (no moving parts)
  → Higher cost and lower capacity compared to HDD.

3) Suitable Usage:
- HDDs: Best for long-term video storage and archives.

- SDDs: Best for real-time recording, buffering and quick access data.

4) Memory Hierarchy and Retrieval speed
Memory hierarchy (Cache → RAM → SSD → HDD) affects speed:

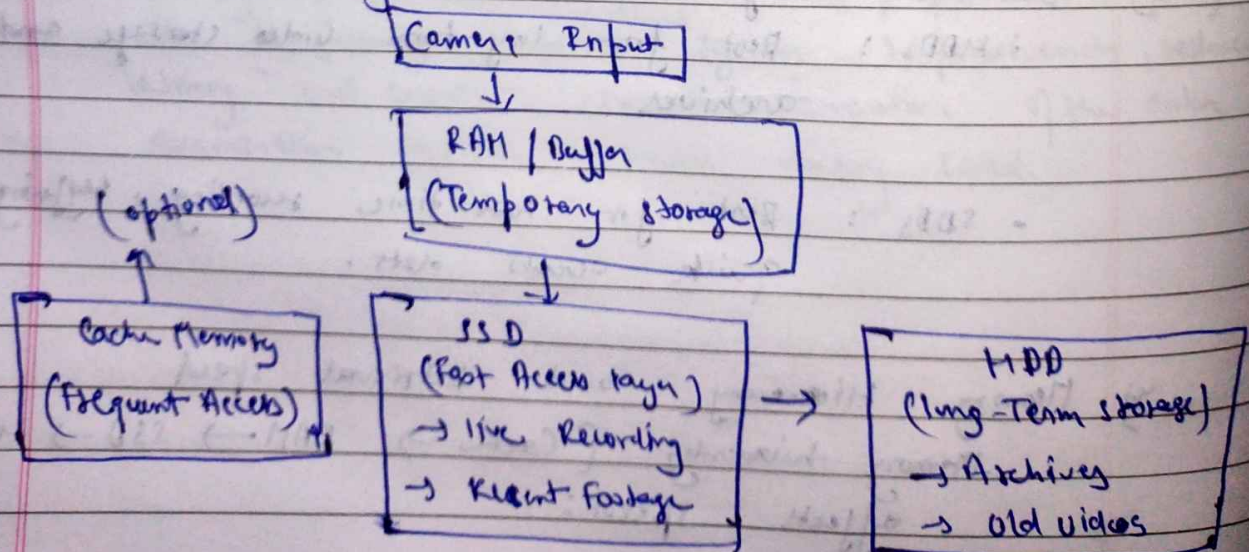
- Data stored in faster memory (cache / SSD) is accessed quickly).
- Data in HDD takes more time.
- Frequently accessed video should be kept in faster storage to improve performance.

5) Optimized Architecture:

Use a hybrid storage system:

- SSD layer: For live recording and ~~recent~~ recent footage
- HDD layer: For long term storage
- Optionally include cache memory for frequently accessed clips.

This balances cost (HDD), speed (SSD), and capacity [HDD + SDD combination]

⇒ Conclusion:

A hybrid architecture using SSDs and HDDs is ideal, ensuring fast access, large storage capacity and cost efficiency for a video surveillance system.